Machine Learning Methods for Finance

Unsupervised Learning - Clustering

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Outline

1. Clustering
   - K-Means
   - Gaussian Mixture Modeling
   - Hierarchical Clustering
   - Minimum Spanning Tree

Motivation

- These notes accompany the Duke Economics course *Machine Learning for Finance*.
- The course focuses on how machine learning methods can be used to design, evaluate, and implement systematic investment strategies.
- Our goal is to introduce a range of ML techniques with enough mathematical structure to build intuition, while avoiding unnecessary formalism.
- Throughout the notes we use a simple return–momentum signal as a running example. Although momentum is not always the ideal illustration for every technique, using a single, familiar signal provides a consistent thread. This is especially true when we transition to real data and empirical implementation.
- The methods we discuss are broadly applicable across many signals, asset classes, and investment settings.

Description of ML Methods

We'll use a toy momentum example as motivation

Unsupervised Learning

- **Clustering**
  Group similar assets by momentum characteristics
- **Dimension Reduction**
  Condense multiple momentum measures into fewer factors
- **Generative Modeling**
  Simulate returns to derive momentum structure

Supervised Learning

- **Classification**
  Label assets as high/low momentum
- **Regression**
  Predict cross-sectional returns using momentum inputs

Time Series Modeling

- **Unsupervised**
  Group assets by temporal behavior, then measure momentum
- **Supervised**
  Forecast returns based on persistence

Defining Momentum

There are many ways to define momentum. Here are two approaches that are not only common, but are designed in a way to support our pedagogical goals.

1. Jegadeesh and Titman: $mom_{i,t} = r_{i,t-12 \to t-2}$
2. Technical Analysis style: $mom_{i,t} = r_{i,t-12} - r_{i,t-2}$
   - Tightly related, and sometimes referred to as "momentum" might be a measure reversal $r_{i,t-2} - r_{i,t-12}$

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Clustering Momentum

Q: How do we form a momentum signal using any clustering method?

1. Convert prices to returns, then compute momentum
2. Group assets by momentum using a clustering method (e.g., K-Means, GMM, hierarchical). Each asset receives a cluster label based on the method's assignment rule.
3. Form a long/short signal by identifying:
   - the cluster with the highest average momentum (the "winner" group), and
   - the cluster with the lowest average momentum (the "loser" group),

   and going long assets in the winner cluster and short assets in the loser cluster.

Types of Clustering

- Hard: Assigns each observation to a single cluster
- Soft: Assigns each observation to potentially more than one cluster with a degree of "belonging"

Clustering Methods

1. Centroid-based — Clusters by grouping observations around representative feature coordinates (e.g., K-Means, K-Medians).
2. Model-based — Clusters by assuming data arise from overlapping probability distributions and assigning observations by likelihood (e.g., Gaussian Mixtures).
3. Hierarchical — Clusters by building a tree of nested groups formed through successive merges or splits.
4. Graph-based — Clusters by connecting observations through a minimum spanning tree and cutting long edges to reveal groups.
5. Density-based — Clusters by identifying dense regions of observations separated by sparse areas (e.g., DBSCAN).

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K-Means

The most common Centroid Based, Hard Clustering method

1. Choose K as the number of clusters
2. Assign each observation x_i to one of the K clusters
3. Compute the mean (aka centroid) of each cluster $\bar{x}^k$
4. For each observation
   - Update cluster assignments by minimizing the distance between each observation and its cluster centroid (i.e. it's cluster "belonging")
   - Update cluster means by recomputing $\bar{x}^k$ for each cluster
   - Repeat until stopping criterion is met

Output

- Cluster means $\bar{x}^k$
- Cluster assignments x_i^k for $k = 1, \cdots, K$; $i = 1, \cdots, N$
- Clustering Error (Within Cluster Sum of Squares) $\frac{1}{N} \sum_{i=1}^{N} (x_i^k - \bar{x}^k)^2$

Considerations for Implementing K-Means

1. Accommodating multidimensional x
2. Choosing the number of clusters
3. Initializing cluster assignments
4. Pre-processing — helpful to standardize, or otherwise scale, the data before clustering, especially when using Euclidean distance
5. Tie Breaking
6. Empty Clusters
7. Stopping Criterion

Note: We won't dwell on 5,6,7 in the list above.

Multidimensional x

Suppose $\mathbf{x}_i^k$ is not a scalar; e.g. Clustering momentum AND volume across the SP500, in which case $dim(\mathbf{x}_i) = 2 \times 1$

- Cluster centroids are columnwise; i.e. (avg momentum, avg volume)
- Clustering Error expands to $\frac{1}{N} \sum_{d=1}^{2} \sum_{i=1}^{N} (x_{i,d}^k - \bar{x}_d^k)^2$

Ticker	Momentum	Errors	Squared Errors
A	1	-1.5	2.25
B	2	-0.5	0.25
C	3	0.5	0.25
D	4	1.5	2.25
Avg	2.5		1.25
Sum			5

Ticker	Volume	Errors	Squared Errors
A	200	-200	40000
B	300	-100	10000
C	700	300	90000
D	400	0	0
Avg	400		35000
Sum			140000

			Error		Squared Errors		
Ticker	Momentum	Volume	Momentum	Volume	Momentum	Volume	Squared Errors
A	1	200	-1.5	-200	2.25	40000	40002.25
B	2	300	-0.5	-100	0.25	10000	10000.25
C	3	700	0.5	300	0.25	90000	90000.25
D	4	400	1.5	0	2.25	0	2.25
Avg	2.5	400					35001.25

Sum of Avg	35001.25
(1/4)*Sum of Sums	35001.25

Note: Euclidean distance is not the only distance measure available for K-Means

Choosing the number of clusters

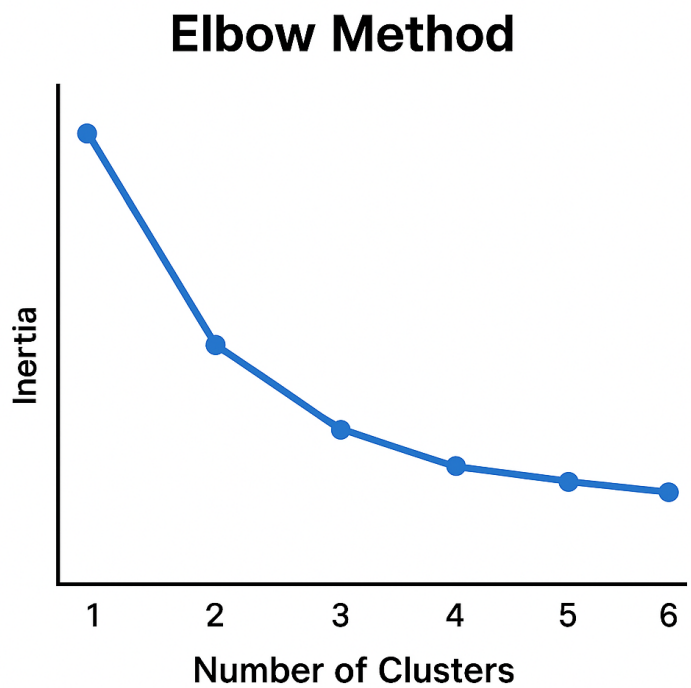
- If we set $K = 1$, then we are basing analysis on the single observation that has the smallest distance measure (e.g. the "average" observation)
- If we set $K = N$, then each observation is its own cluster

Typical Approaches

1. Choose K based on context; e.g. Long, Neutral, Short $K = 3$
2. If clustering is an aspect of our feature engineering to a supervised ML exercise, we can chose the K that optimizes the accompanying supervised validation error
3. If clustering is primarily part of EDA, then we can use "elbow method". Plot the within Cluster Error (aka Inertia) for various k and chose the K where the rate of decrease in the error slows.

Choosing Number of Clusters with Elbow Method

Q: How many clusters are suggested?



A: Elbow plots are a crude, but useful, heuristic which requires a degree of subjectivity. In this Elbow plot, it appears that 4 clusters are appropriate.

Initializing Cluster Assignments

A few of the many approaches available

- Randomly, assuming a uniform distribution, choose rows from the dataset as the initial centroids.
- (K-means ++) Ensure the initial centroids are well spread out
  1. Pick first centroid randomly
  2. Compute squared distances of each observation to that centroid
  3. Choose next centroid with probability proportional to the squared distance so that points farther away are more likely to be chosen
  4. Repeat until K centroids are chosen.

Judging Cluster Assignment with Silhouette Plots

Silhouette Score

$$s_i = \frac{b_i - a_i}{max\{a_i, b_i\}} \in \{-1, \cdots, +1\}$$

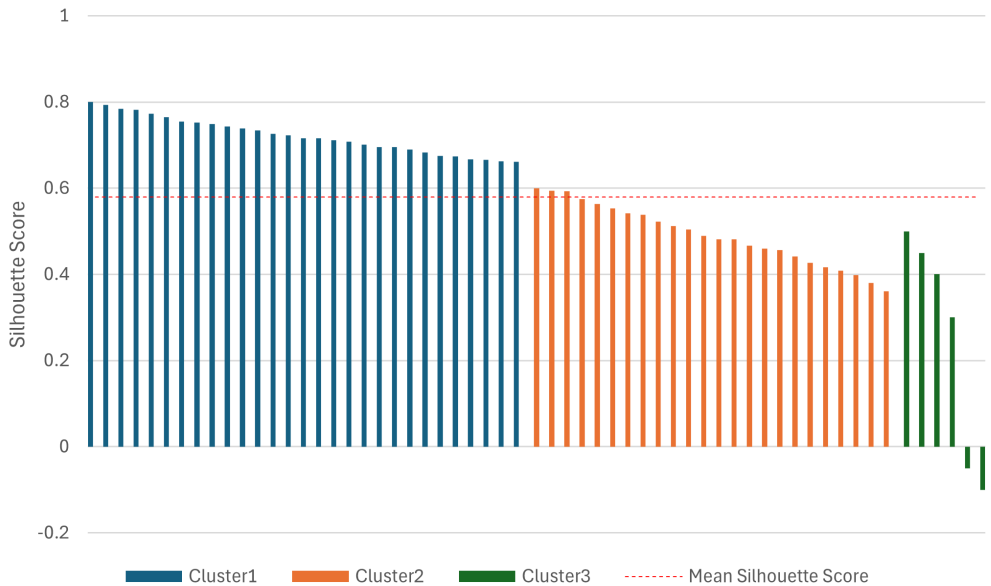
where

- a_i : cluster cohesion; average distance of observation i to all other points in its own cluster
- b_i : cluster separation; average distance between observation i and all observations in the nearest cluster (to which i is not assigned)

We compute the mean Silhouette score $\bar{s}$ and compare the individual scores to that mean. If the scores within a cluster are less than the $\bar{s}$, it implies that this cluster might be too dispersed (large a_i) and/or too close to other clusters (small b_i).

Good and Bad Clusters

Q: Which cluster(s) are good/bad? What does a negative score mean?



A:

- Cluster 3 seems "bad". Many of the observations assigned to this cluster have lower scores than average.
- The negative scores imply that these observations are actually closer to another cluster than to the cluster to which they are assigned.

K-Means Clustering Momentum

Q: How form a momentum signal via K-Means Clustering?

1. Choose the number of clusters K via Elbow, Silhouette, etc.
2. Group assets by momentum via **K-Means** such that the cluster assignment for asset i at time t is the cluster whose centroid is closest to its momentum value

$$s_{i,t} = \mathbf{1}\Big[p^{(+)}(X_{i,t}, Z_t)\Big] - \mathbf{1}\Big[p^{(-)}(X_{i,t}, Z_t)\Big]$$
$$p^{(+)} = \mathbf{1}\Big[k_t^i = k_t^{(+)}\Big] \quad p^{(-)} = \mathbf{1}\Big[k_t^i = k_t^{(-)}\Big]$$

where

- $k_t^i = \arg\min_k \|X_{i,t} - \mu_{k,t}\|$ (cluster label for asset i)
- $\mu_{k,t} = \frac{1}{N_{k,t}} \sum_{i:\, k_t^i = k} X_{i,t}$ (centroid of cluster k at time t)
- $\bar{m}_{k,t} = \frac{1}{N_{k,t}} \sum_{i:\, k_t^i = k} mom_{i,t}$ (average momentum in cluster k)
- $k_t^{(+)} = \arg\max_k \bar{m}_{k,t}$, $k_t^{(-)} = \arg\min_k \bar{m}_{k,t}$ (highest- and lowest-momentum clusters)
- $mom_{i,t}$: momentum of asset i
- $X_{i,t}$: momentum feature for asset i; Z_t: momentum vector for all other

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Motivating Gaussian Mixture Models for Momentum

Common application of Centroid Based, Soft Clustering

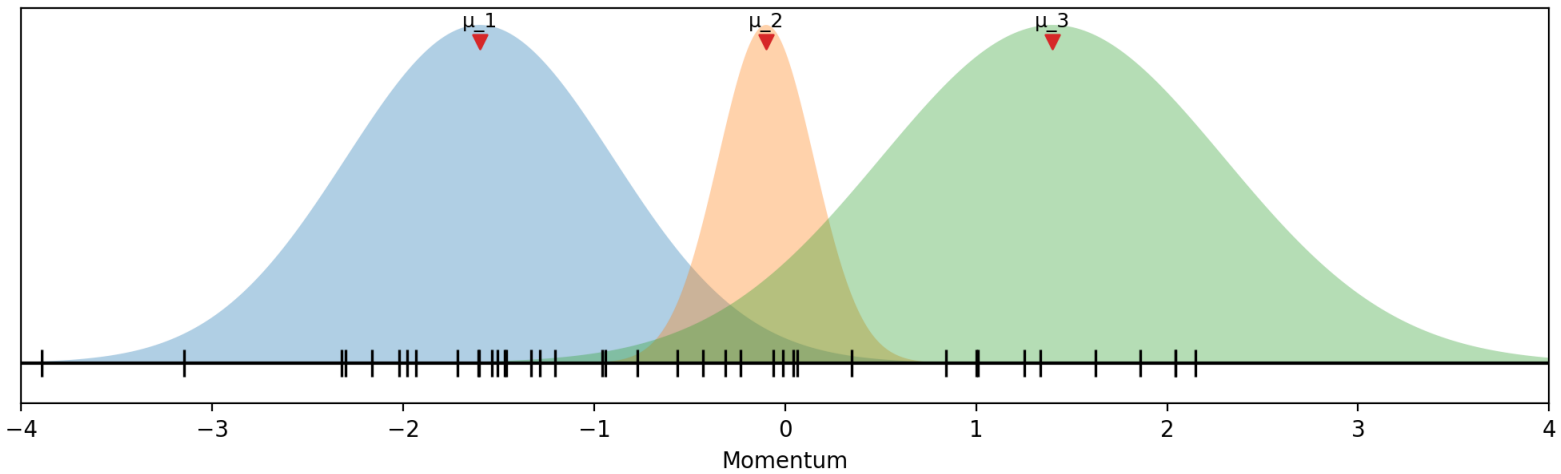
- Consider Momentum as a *continuous, noisy* signal
- There are no natural hard cutoffs between:
  - Losers
  - Neutral assets
  - Winners
- Assets near the boundary may plausibly belong to multiple regimes

GMM perspective:

- Observed momentum is generated by latent momentum regimes
- Each regime has its own mean and variance
- Each asset has a *probabilistic* cluster assignment

Motivating Gaussian Mixture Models (GMM)

Common application of Centroid Based, Soft Clustering



- Consider each bar on the horizontal axis to be an asset
- The asset near -4 is clearly in cluster 1
- The assets near 2 are clearly in cluster 3
- But the assets between -1 and 0.5 appear to have reasonable probabilities of being in multiple clusters

Steps for GMM Clustering

1. Initialize number of clusters K, cluster means $\bar{x}^k$, covariances Σ_x^k, and prior probabilities p^k that i is a member of cluster k
2. Update "degrees of belonging" (posterior probabilities)

$$\hat{p}(k^i = k \mid x_i) \propto \underbrace{\hat{p}^k}_{\text{prior}} \times \underbrace{N(x_i; \hat{\bar{x}}^k, \hat{\Sigma}^k)}_{\text{likelihood}}$$

3. For each cluster k update
   - Cluster probability: $\hat{p}^k = (N_k/N)$ with $N_k = \sum_{i=1}^N \hat{p}(k^i = k|x)$
   - Cluster mean: $\hat{\bar{x}}^k = (1/N_k) \sum_{i=1}^N \hat{p}(k^i = k|x) \times \mathbf{x}_i$
   - Cluster covar: $\hat{\Sigma}_x^k = (1/N_k) \sum_{i=1}^N \hat{p}(k^i = k|x) \times (\mathbf{x}_i - \hat{\bar{x}}^k)(\mathbf{x}_i - \hat{\bar{x}}^k)'$
   - Repeat until convergence

Output: predicted "degrees of belonging" $\hat{p}(k^i = k|x)$; i.e. probability that asset i is a member of each of the K clusters

GMM Clustering Momentum

Q: How form a momentum signal via GMM Clustering?

1. Choose the number of clusters K via Elbow, Silhouette, etc.
2. Group assets by momentum via **GMM with a chosen threshold of cluster belonging** such that the cluster assignment for asset i at time t is the cluster for which the degree of belonging is the highest $k_t^i = k|max(\hat{p})$

$$s_{i,t} = \mathbf{1}\Big[p^{(+)}(X_{i,t}, Z_t)\Big] - \mathbf{1}\Big[p^{(-)}(X_{i,t}, Z_t)\Big]$$

$$p^{(+)} = \mathbf{1}\Big[k_t^i = k_t^{(+)}\Big] \quad p^{(-)} = \mathbf{1}\Big[k_t^i = k_t^{(-)}\Big]$$

where

- $k_t^{(+)} = \arg\max_k \bar{m}_{k,t}$; $k_t^{(-)} = \arg\min_k \bar{m}_{k,t}$
- X_{it} = momentum for asset i; Z_t = momentum for other assets
- $\bar{m}_{k,t} = \frac{1}{N_{k,t}} \sum_{i:\, k_t^i = k} mom_{i,t}$
- $mom_{i,t}$: momentum of asset i
- k_t^i: cluster label
- $\bar{m}_{k,t}$: average cluster-level momentum

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Hierarchical Clustering

Hierarchical clustering groups data within a tree structure based on similarity

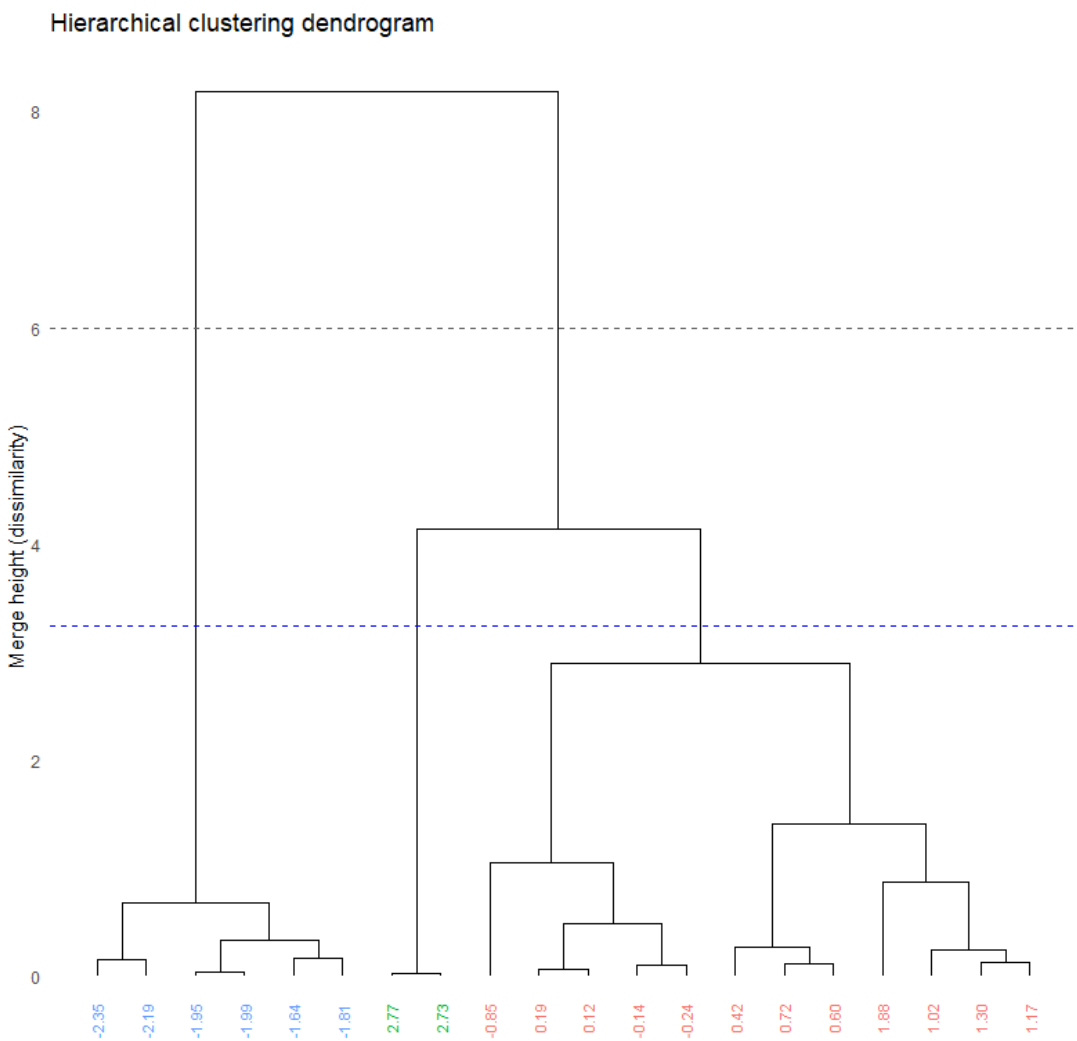
- Consider Momentum as a relative signal
- Assets may not be attractive or unattractive in isolation
- They are attractive relative to their peer group at a point in time

Hierarchical Clustering perspective:

- Arrange assets by similarity, not by arbitrary cutoff, as in K-Means
- Produce a ranking structure, not only clusters
- Choose the number of clusters *after* seeing the data ordering

Hierarchical Clustering - Dendrogram

Suppose the horizontal axis measures momentum, and each observation is another asset



Q: Why is the vertical line for 2.77 and 2.73 so large?
A: Very different from the other assets.
Q: How many clusters chosen with the blue horizontal line?
A: 3

Q: How many clusters chosen with the gray horizontal line?
A: 2; the green and red are grouped together.

Ways to create the hierarchy

- Agglomerative: Start with each observation as a cluster and merge the closest pairs until a single cluster remains (typical)
- Divisive: Start with a single cluster and iteratively split into smaller clusters until each cluster contains only one observation

Steps for Agglomerative Clustering

1. Compute a pair-wise distance matrix of all observations e.g. $d(i,j) = (x_i - x_j)^2$
2. Initialize with one cluster per observation e.g. N clusters; one for each asset
3. Merge the two closest clusters based on either
   - Single linkage: distance = min pairwise distance between members
   - Complete linkage: distance = max pairwise distance between members
   - Average linkage: distance = average pairwise distance
   - Ward's method: merge clusters that minimize total within-cluster variance
4. Update distance matrix to reflect the new cluster
5. Repeat until one cluster remains

Adapting Hierarchical Clustering for Multidimensional x

Suppose we are clustering on 12mth and 2mth returns such that $x_i = (12mth, 2mth)$

1. Compute a pair-wise distance matrix of all observations e.g., $d(i,j) = (1/2)[(12mth_i - 12mth_j)^2 + (2mth_i - 2mth_j)^2]$
2. Initialize with one cluster per observation e.g. N clusters; one for each asset
3. Merge the two closest clusters. Ward's method (minimize total within-cluster variance) typically preferred
4. Update distance matrix to reflect the new cluster
5. Repeat until one cluster remains

Hierarchical Clustering Momentum

Q: How form a momentum signal via Hierarchical Clustering?

1. Choose the number of clusters via Dendrogram
2. Assign each asset to a momentum cluster using hierarchical clustering with a chosen dissimilarity cutoff, producing a deterministic cluster label k_t^i

$$s_{i,t} = \mathbf{1}\left[p^{(+)}(X_{i,t}, Z_t)\right] - \mathbf{1}\left[p^{(-)}(X_{i,t}, Z_t)\right]$$

$$p^{(+)} = \mathbf{1}\left[k_t^i = k_t^{(+)}\right] \quad p^{(-)} = \mathbf{1}\left[k_t^i = k_t^{(-)}\right]$$

where

- $k_t^{(+)} = \arg\max_k \bar{m}_{k,t}$; $k_t^{(-)} = \arg\min_k \bar{m}_{k,t}$
- X_{it} = momentum for asset i; Z_t = momentum for other assets
- $\bar{m}_{k,t} = \frac{1}{N_{k,t}} \sum_{i:\, k_t^i = k} mom_{i,t}$
- $mom_{i,t}$: momentum of asset i
- k_t^i: cluster label
- $\bar{m}_{k,t}$: average cluster-level momentum

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Minimum Spanning Tree (MST)

MST connects all observations using only pairwise distances, choosing edges so that the total connection cost is minimized

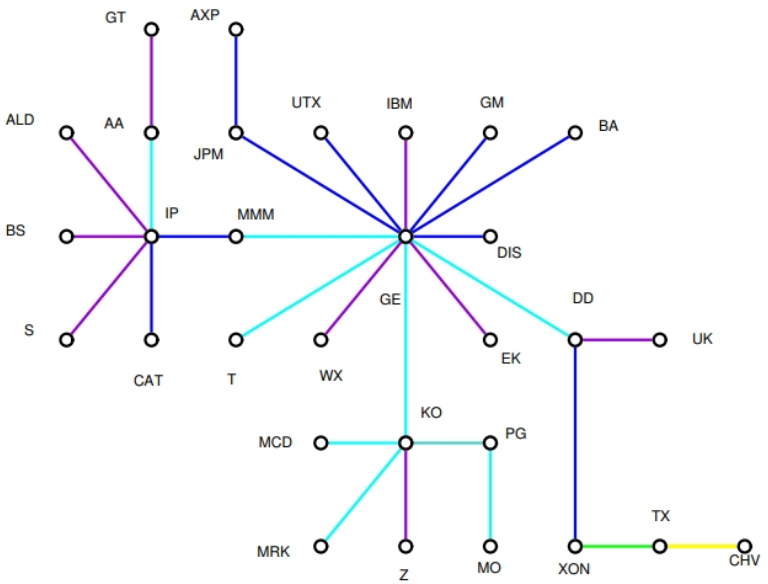


Figure: Mantegna, R. N. *Hierarchical Structure in Financial Markets*. arXiv:cond-mat/9802256 (1998)

- Short edges reflect strong positive correlation. Assets connected by short links move similarly and form local clusters.
- Long edges reflect weak similarity — low or mildly negative correlation. Assets with strong negative correlations i) likely don't have an edge connecting them since the MST picks only the shortest edges, and ii) likely appear on opposite sides of the tree.
- Longer paths between two assets indicate weaker overall relation. If two assets connect only through many intermediate nodes, their direct correlation was not strong enough to appear as a short or direct edge.

Building an MST from Return Correlations

1. Compute asset returns (e.g., daily log returns)
2. Compute the return correlation matrix
3. Convert correlations to a distance* metric:

$$d_{ij} = \sqrt{2(1 - \rho_{ij})} \in [0, 2]$$

4. Construct the Minimum Spanning Tree (MST) from the distance matrix
   1. Initialize with any asset (e.g. Prim's algorithm)
   2. Iteratively add the asset whose distance to the existing tree is smallest (i.e., most highly correlated with at least one asset already in the tree)
   3. Continue until all assets are connected by $N - 1$ edges

*: Distance metrics satisfy properties i) $d_{ii} = 0$, ii) $d_{ij} \geq 0$, iii) $d_{ij} = d_{ji}$, iv) Triangle Inequality ($d_{ij} \leq d_{ik} + d_{kj}$)

Measuring Centrality

Centrality quantifies how much a network is concentrated around a single asset

Here are a few common centrality measures for asset (i):

1. Degree Centrality: # of edges (Number of connections)
   - Q: Why shouldn't we use this centrality measure for asset returns?
   - A: All assets tend to be connected so the centrality will typically be the same for all assets
2. Weighted Degree Centrality: $\sum w_{ij}$ (Strength of connections), which captures both the number and strength of connections while treating all neighbors as equally important
3. Eigenvector Centrality: (Systematic importance; i.e. Strength of connections to other important assets)
   - An asset is important if it is connected to other important assets
   - Let $W = (w_{ij})$ be a similarity (weight) matrix
   - Solve the eigenvalue problem $Wc = \lambda c$
   - Normalize so that $\sum_i c_i = 1$
   - Centrality score $c_i = \frac{1}{\lambda} \sum_{j=1}^{N} w_{ij} c_j$

Clustering With MST

1. Form the MST, which produces nodes (observations), edges (connections), and weights (strength of connections measured by distance metric)
2. Remove the $k-1$ largest edges. k serves as the number of clusters, which can be determined via Elbow, Silhouette, etc..
3. The remaining graph is a forest; i.e. a set of trees, rather than a single tree, where each tree is a cluster

MST Clustering Momentum

Q: How form a momentum signal via MST Clustering?

1. Choose the number of clusters K via network graph
2. Group assets by momentum via **MST**, where the cluster assignment for asset i at time t is the connected component (tree) containing asset i after edge removal

$$s_{i,t} = \mathbf{1}\left[p^{(+)}(X_{i,t}, Z_t)\right] - \mathbf{1}\left[p^{(-)}(X_{i,t}, Z_t)\right]$$

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